UNIVERSITY OF LAY ADVENTISTS OF KIGALI

UNILAK RWAMAGANA CAMPUS

IDFinder Rwanda

Lost and Found Identity Document Recovery System

Milestone 1

Problem Discovery and User Research

Group 2

MISIGARO Thomas — M04930/2025

SIBOBUGINGO Leonidas — M04070/2025

Lecturer Yves Himbaza Mugiraneza

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1 Introduction and Project Background

IDFinder Rwanda is a proposed system for supporting the recovery of lost national identity cards and driving licences in Rwanda. The project focuses on the period between losing an original document and learning whether another person has found it. During the informal discussions reported by the project team, people described recovering a lost identity document as difficult. This concern provided the starting point for examining how document owners search, how finders report a discovery and how a document can be returned through an appropriate handover process.

Milestone 1 examines the problem before detailed software requirements or implementation are developed. Following the lecturer's discovery approach, the report considers the people affected, the existing recovery routes, the information needed during a search and the uncertainty that can occur at each stage. The purpose is to establish a clear account of the recovery problem and identify questions that user research must answer. The proposed system direction therefore remains connected to the discovery findings and the limits of the available evidence.

The evidence available for this report consists of the team's account of informal discussions, the three illustrative dialogues retained from the presentation and published information about existing services. The original presentation explicitly describes the dialogues as fictional walkthroughs. Their responses are reproduced and analysed as scenarios in this report. They help explain possible experiences, while the informal discussions support the broader observation that recovery can be difficult. No interview dates, participant counts, search durations or recovery rates were recorded in the material supplied.

1.1 Project Team and Responsibilities

MISIGARO Thomas, registration number M04930/2025, holds the Engineering Lead role. His responsibilities include organising technical discovery records, maintaining the project repository and preparing the transition from the problem statement to testable requirements. SIBOBUGINGO Leonidas, registration number M04070/2025, holds the UX and Product Lead role. His responsibilities include preparing the interview instrument, organising user needs and documenting the proposed user experience. Both team members contribute to research, analysis and explanation of the complete project, consistent with the course requirement for shared understanding.

2 Assessment of the Selected Problem

The first discovery criterion is whether the problem is real. The team reported that the people consulted considered it difficult to recover a lost identity document. This is a relevant initial indication because it concerns an experience the proposed users recognise. However, the discussions were not preserved as individual interview records. The present evidence establishes an initial concern and supports further investigation, while the detailed causes still require confirmation through documented incidents.

The frequency criterion concerns how often the difficulty occurs and how many people experience it. The available discussions did not measure frequency. It would therefore be inappropriate to assign a percentage, estimate a national total or state that every document owner encounters the same obstacle. The report instead identifies the point at which frequency needs to be investigated, namely whether owners repeatedly check places and contacts because they cannot determine whether a found document has already been reported.

The pain criterion concerns the consequences for the person involved. The scenarios describe returning to previous locations, asking several people and repeating an explanation. These actions illustrate how a search could require effort, calls or trips. They also show that uncertainty may continue even after a finder has handed a document to someone nearby. The material contains no measured financial cost or elapsed time, so these consequences are treated as possible burdens to investigate.

The reachability criterion concerns whether the team can access suitable users. The proposed research would involve people who have recently lost a national ID or driving licence and people who have found one. A receiving worker may also explain how documents are held and collected. The buildability criterion is addressed by limiting the opportunity to reporting, searching and a controlled handover process. This scope is small enough to inform a course project, subject to requirements validation and approval.

3 Discovery Questions and Current Recovery Journey

Five questions organise the discovery. The first asks where an owner looked immediately after noticing a loss. The second asks whom the owner contacted and why that person or channel was selected. The third asks what a finder did with the document and how they decided where to leave it. The fourth asks how the owner and finder could tell whether a possible match was genuine. The fifth asks how either person learned that the document had

been collected. Together these questions follow an incident from the first search through to its outcome.

A possible recovery journey begins when an owner notices that a document is missing and returns to a place where it was last used. The owner may then ask familiar contacts or check a receiving location. Separately, a finder may discover the card and leave it with a worker or an appropriate receiving office. The central discovery issue is whether those two journeys become connected. If the owner checks one channel while the document is held elsewhere, the original may remain available without the owner knowing where to find it.

Published guidance demonstrates that existing recovery and replacement routes already form part of this context. Rwanda National Police advises that a found driving licence should be returned to a known owner or handed to the nearest police station (Rwanda National Police, 2022). IremboGov describes services for replacing a lost national ID and obtaining a duplicate driving licence (IremboGov, n.d.-a, n.d.-b). These sources establish relevant existing services. They do not establish the frequency or duration of the particular difficulties described in the project scenarios.

4 Stakeholder Identification and Analysis

The stakeholder analysis considers people who directly use the recovery process and organisations whose responsibilities influence it. Document owners have a strong interest because they need to determine whether an original document has been found. Finders also have a direct interest because they need a clear and appropriate route for reporting or handing over a discovery. Receiving staff affect the physical custody of the document and the record of collection. These groups provide different accounts of the same journey and should be represented in discovery.

The wider stakeholder group includes the National Identification Agency, Rwanda National Police, data protection oversight bodies, local administrative offices and organisations where documents may be found or held. NIDA and the police have responsibilities connected with identification and driving documentation. Local offices and receiving organisations may influence where a person asks for help. Data protection stakeholders influence how personal information should be handled. Listing these stakeholders identifies potential constraints and consultation needs. It does not imply that any institution has agreed to participate in the pilot.



Figure 1 Stakeholder power and interest map retained from the original presentation

The power and interest map helps the team decide how each stakeholder should be involved. Direct users need opportunities to explain their experiences. Institutions with authority over documents or related procedures need their responsibilities considered when the project scope is defined. The immediate discovery priority remains the owner and finder, supported by information about receiving and collection practices. Their experience is necessary to decide whether a shared recovery record would address a meaningful difficulty.

5 Brainstorming and Opportunity Exploration

The brainstorming exercise examined four possible actions involved in recovering an original document. A person may search by area and date, route a found document to an appropriate collection point, track a private claim or verify and record a handover. These actions describe different stages of recovery rather than separate promises about the final application. Considering them together helps the team examine whether the proposed process follows the needs of both the owner and the finder.

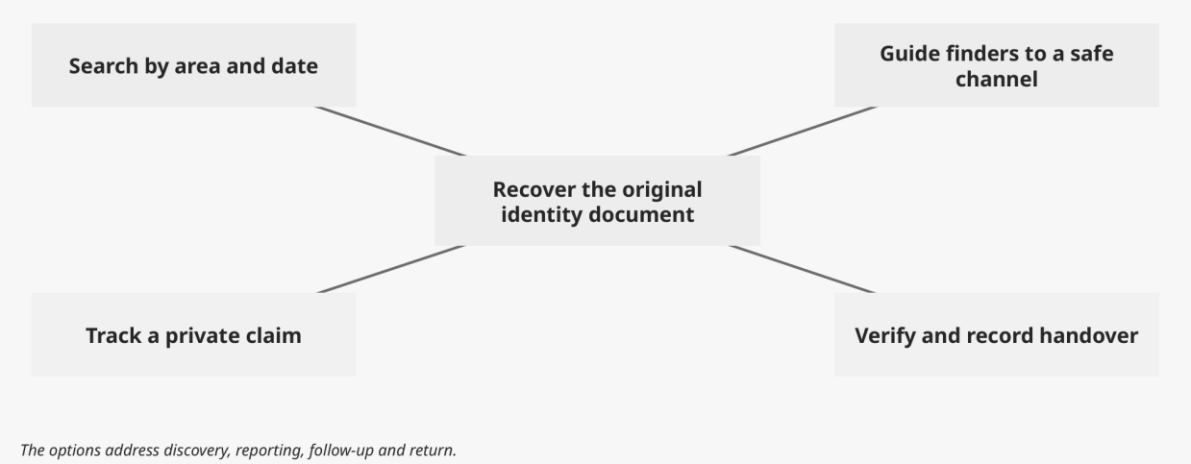


Figure 2 Brainstorming board retained from the original presentation

The initial opportunity is to make the connection between discovery and return easier to follow. A report could describe the document without publishing unnecessary personal information, while an owner could look for a possible match and make a private claim. Before this direction becomes a requirement, the team must determine who would receive the document, who would verify ownership and how the outcome would be recorded. These decisions depend on the operating context and the availability of a suitable collection arrangement.

6 User Research Approach

The interview approach follows the lecturer's emphasis on past behaviour. Participants would be asked to describe a particular incident, beginning with what happened and continuing through the actions taken and the final outcome. Follow-up questions such as "Why?" and "Tell me more" are used to explore a decision or difficulty in the account. This approach is intended to uncover actual search and handover practices before asking participants to evaluate a proposed application.

The screening instrument first checks whether the person is willing to participate, whether notes may be taken and whether the incident concerns a lost or found national ID or driving licence. The incident section records what happened, where the search or reporting began, the contacts involved, the hardest step and the eventual outcome. A short open-ended explanation is more informative than an answer limited to whether the person likes the project idea. It gives the team a sequence that can be compared across accounts.

Participation should be voluntary, and a person should be able to decline a question or end the conversation. Research records should use participant codes and avoid unnecessary document numbers, photographs or private contact information. Permission to take notes should be obtained before recording the discussion. The supplied material does not document completed consent procedures or recorded interviews. Accordingly, this section describes the research approach and the instrument, while the following section preserves the original illustrative responses.

7 Interview Questions and Illustrative Responses

7.1 Scenario P1 National ID Owner

MISIGARO Thomas facilitated the national ID owner walkthrough in the original presentation. The opening question was, “When did you realise the ID was missing?” The illustrative response was, “I noticed when I needed to show it.” This answer places the start of the search at the moment the document was needed. It suggests an interview follow-up about what the person attempted immediately afterwards and whether the need to use the document affected their decisions.

The next question was, “Where did you look first? Why?” The response was, “I returned to the last place I remembered using it.” When asked, “Tell me more about what happened there,” the owner responded, “I asked staff whether anyone had left it with them.” The two answers illustrate a search organised around a remembered location and the knowledge of people at that location. They identify a possible dependence on individual enquiries rather than a record the owner can check directly.

The final question was, “How did the search end?” The response was, “I checked nearby but could not tell whether it had been found.” The main design concern illustrated here is uncertainty about the outcome. The scenario does not establish that the document was never recovered. It shows that the owner lacked confirmation at the point described. That distinction matters when considering whether the system should help people discover reports, track claims or both.

7.2 Scenario P2 Driving Licence Owner

SIBOBUGINGO Leonidas facilitated the driving licence owner walkthrough. The first question was, “What happened the last time you lost your licence?” The illustrative response

was, “I noticed after returning from a trip.” The question then moved to the first contact by asking, “Who did you contact first? Why?” The response was, “The people I travelled with knew where I had stopped.” This sequence illustrates a search in which the owner depends on other people to reconstruct possible locations.

The follow-up question was, “Tell me more about the hardest step.” The response was, “I had to repeat the story to each new person I asked.” This statement illustrates the effort associated with making several separate enquiries. It suggests that the same incident details may have to be explained repeatedly when there is no shared record available to the people involved. The account remains a scenario, so it cannot be used to calculate how often this occurs among licence holders.

The closing question was, “What did you do in the end?” The response was, “I stopped searching and considered applying for a replacement.” The answer identifies a possible transition from recovery to replacement. It does not show that replacement was completed or that searching had become objectively impossible. It illustrates why the project should consider how an owner decides when further recovery enquiries are worthwhile and how the proposed service can avoid giving false certainty.

7.3 Scenario P3 Finder of an Identity Document

MISIGARO Thomas facilitated the finder walkthrough. The opening question was, “Tell me about the last time you found an ID.” The illustrative response was, “I found a card near a shop.” The next question asked, “What did you do first? Why?” The finder responded, “I asked the shop worker because the owner might return there.” This sequence illustrates a locally reasonable decision based on the possibility that the owner would retrace their steps.

The question, “Tell me more about what you shared,” received the response, “I described the card without posting its number.” This introduces a privacy concern into the recovery journey. A finder may need to make the discovery visible while avoiding disclosure of information that is unnecessary for a public notice. The scenario supports asking what details an owner needs to recognise a possible match and what information should be reserved for a private verification process.

The final question was, “How did you learn the outcome?” The response was, “I left it with the worker and did not hear what happened next.” The illustration highlights a gap in follow-up after physical handover. The finder may have acted helpfully but still lack confirmation of

return. This creates a separate question from initial reporting, namely whether a completed recovery should produce an outcome record visible to the appropriate people.

8 Affinity Analysis and Five Candidate Pain Points

The affinity exercise groups related observations under search, reporting, effort, privacy and follow-up. The original map distinguishes the broad concern reported through informal discussion from the themes explored in the scenarios. This distinction is retained because a well-organised map does not increase the strength of the underlying evidence. Each theme provides a candidate problem to check against documented user accounts.

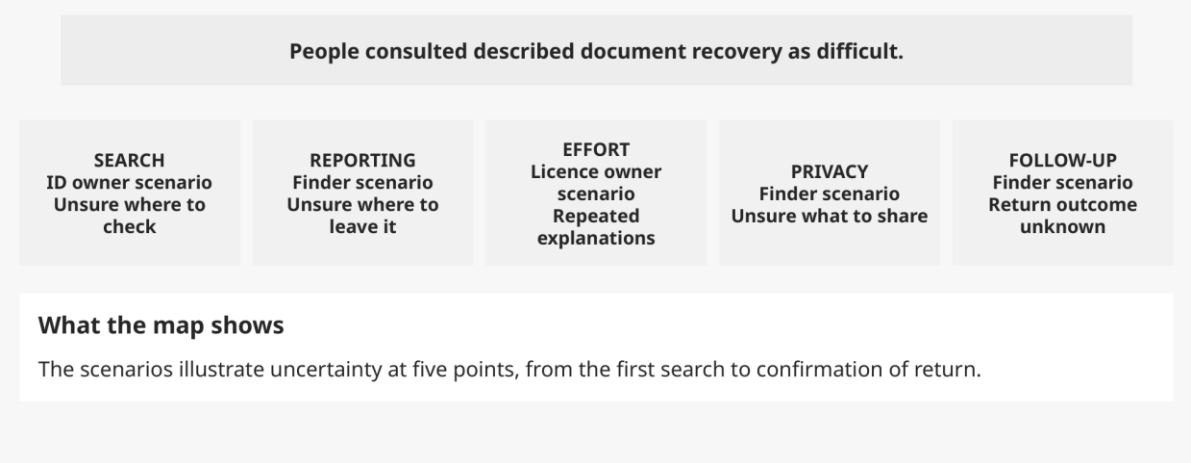


Figure 3 Affinity map retained from the original presentation

8.1 Uncertainty About Where to Search

The first candidate pain point is uncertainty about where to look for a found document. P1's illustrative statement, "I checked nearby but could not tell whether it had been found," expresses the gap between making enquiries and knowing the outcome. A possible design implication is a searchable report record with enough information to identify a potential match. The discovery work must still establish which locations and channels owners currently use.

8.2 Uncertainty About Where to Report a Discovery

The second candidate pain point concerns the finder's reporting decision. P3 stated, "I asked the shop worker because the owner might return there." The answer illustrates how a finder can select a handover location based on expectation rather than a shared reporting process. The implication is to explore a clear receiving route and reliable custody arrangements. A

local handover and an online report must remain connected so that a possible match corresponds to a document that can actually be collected.

8.3 Repeated Explanations During the Search

The third candidate pain point is repeated effort. P2 explained, “I had to repeat the story to each new person I asked.” The statement illustrates how a fragmented search may require the owner to start again with each contact. A structured record could potentially reduce repeated explanation by retaining consistent incident details. Whether this would make a meaningful difference should be investigated through actual examples of repeated enquiries.

8.4 Uncertainty About Safe Information Sharing

The fourth candidate pain point concerns what a finder should disclose. P3’s statement, “I described the card without posting its number,” illustrates a choice to limit public information. This supports examining a distinction between public search information and private ownership information. The project should establish the minimum information needed for a possible match and the responsibilities of the person or organisation handling a claim.

8.5 Lack of Confirmation After Handover

The fifth candidate pain point is the absence of a clear outcome after a document is left with someone else. P3 stated, “I left it with the worker and did not hear what happened next.” A recorded handover could help authorised participants understand whether a recovery is still pending or completed. The implication is a need to investigate who can confirm collection and what evidence should support that confirmation.

9 Provisional User Profiles

9.1 Document Owner

The document owner profile represents the needs illustrated by P1 and P2. The person’s main goal is to learn whether the original national ID or driving licence has been found and, where possible, recover it through an understandable process. The scenarios show an owner returning to familiar places, asking contacts and trying to decide whether to continue searching. The central frustration is uncertainty about the location and status of the document.

This provisional profile suggests a need for a clear search route, understandable descriptions of possible matches and a way to follow a claim. It does not assign an invented name, age, occupation, device type or level of digital literacy. Such characteristics would require evidence from real participants. The design should therefore allow research findings to refine the profile before interface decisions are fixed.

9.2 Finder

The finder profile represents the needs illustrated by P3. The finder's goal is to return a document through an appropriate channel while sharing only the information needed. The example finder approached a worker near the discovery location, limited the details shared and did not learn the eventual outcome. This profile therefore connects reporting, custody and confirmation of return.

The proposed service needs to be understandable to a person whose involvement may end soon after making a report. It should make the next step clear and explain what happens to the report after submission. The profile remains provisional because the supplied evidence does not establish common reporting preferences, access to a smartphone or willingness to maintain an account.

10 Existing Solutions and Comparative Analysis

10.1 Physical Handover Through a Police Station

Rwanda National Police provides an existing route relevant to found driving licences. Its published advice tells a finder to return a licence to a known owner or hand it to the nearest police station (Rwanda National Police, 2022). The strength of this route is that a physical document can be transferred to an identifiable receiving institution. For IDFinder, the comparison raises the question of how the owner would discover where a document has been handed in. The source does not establish the presence or absence of a nationwide searchable recovery database.

10.2 IremboGov National ID Replacement

IremboGov describes a national ID replacement service for people whose cards are lost or damaged (IremboGov, n.d.-a). This is an established administrative route for addressing the need for a replacement document. Its purpose differs from finding an original card that

another person may already be holding. The comparison helps define the proposed project boundary by focusing IDFinder on discovering a possible original and organising a return, while recognising that official replacement remains a relevant option for an owner.

10.3 IremboGov Duplicate Driving Licence

IremboGov also describes a duplicate driving licence service for lost or damaged definitive licences (IremboGov, n.d.-b). This route addresses a specific document category and provides a defined application process. The discovery opportunity concerns the earlier stage in which an owner may still be searching for the original licence. The proposed system would need to make its scope clear so that a recovery report is understood as information about a possible found document.

Community messaging remains an additional route considered in the original presentation. A message can alert contacts who may have useful local information, but the team has no measured evidence of its reach, reliability or follow-up. Compared with the three documented routes above, it illustrates the possibility of dispersed information. The comparison supports investigating how reporting, search and collection could be connected without assuming that existing services have failed.

11 Problem Statement

People in Rwanda who lose a national ID or driving licence may struggle to recover the original document because the information available to owners, finders and receiving locations may be difficult to connect. In the scenarios, owners relied on remembered locations and separate enquiries, while the finder left a document locally and did not learn the outcome. These examples suggest possible uncertainty about search, reporting, claim handling and confirmation of return.

The team's informal discussions support the broad concern that recovery can be difficult. Published guidance confirms the relevance of physical handover and official replacement routes. However, the specific causes illustrated by the dialogues have not been verified through documented interviews. The working problem statement therefore identifies a recovery coordination problem for investigation, with possible consequences including extra effort, repeated explanations and uncertainty about whether a document remains recoverable.

12 Proposed Solution Direction and Pilot Boundary

The proposed direction is a limited reporting and recovery platform. A finder would submit a report containing the information needed to describe the discovery. An owner would check for a possible match and submit a private claim. A confirmed collection arrangement would support ownership verification and a recorded handover. The intended value is a clearer connection between a found report and the eventual return of the physical document.

The initial scope covers national IDs and driving licences. Publicly visible information would be limited, while claim details would be handled privately. The operating process would need an identified receiving location and a person responsible for verification and collection records. These are assumptions that must be validated before the next milestone fixes the requirements. The project does not establish an agreement with a public institution or access to an official identification database.

13 Discovery Conclusion

Milestone 1 has organised the original idea around the owner's search and the finder's handover experience. The stakeholder analysis, brainstorming and affinity map show how the project could investigate the connection between reporting and return. The three illustrative dialogues provide detailed examples of the questions to ask and the possible difficulties to explore. The existing solution comparison places that opportunity alongside physical handover and official replacement services.

The available evidence supports continued investigation of recovery difficulty, while the five detailed pain points and the two user profiles remain provisional. Documented interviews are needed to establish which difficulties occur in real incidents and which should shape the pilot. The next milestone can then translate confirmed needs into requirements that identify what the system must do, who is authorised to do it and how successful recovery will be recognised.

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